

Vedang Kshirsagar

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LinkedIn — Webpage

Experience

AIRA Matrix

AI Engineer

Mumbai, India

2024 – Present

- Developed deep learning models for medical image analysis and histopathology applications.
- Designed and evaluated machine learning pipelines for classification, restoration, and anomaly detection.
- Conducted experiments involving multiple instance learning, generative models, and explainable AI.

Medical and Environmental Computing Lab, TU Darmstadt

Research Intern | Advisor: [Dr. Anirban](#) | [\[Report\]](#)

Darmstadt, Germany

May 2023 – July 2023

- Explored robust 3D cerebral ventricle segmentation under intra-ventricular hemorrhage (IVH), addressing the challenging problem of blood-induced boundary ambiguity in clinical CT volumes.
- Developed a three-stage, dual-U-Net refinement pipeline within the self-configuring nnU-Net framework for progressive segmentation of ventricles and hemorrhagic regions.
- Evaluated the proposed architecture through 5-fold cross-validation, investigating gradient detachment, multi-stage loss balancing, and anatomical symmetry augmentation while benchmarking against a strong nnU-Net baseline.

Education

Indian Institute of Technology (BHU), Varanasi

B.Tech. + M.Tech. in Engineering Physics

CGPA: **9.04** / 10

Varanasi, India

2019 – 2024

Publications

• Learning Structure-Aware Foundational Representation of Rat Testicular Tubules Using Multiple Instance Learning

Vedang Kshirsagar *, Saketh Juturu *, Geetank Raipuria, and Nitin Singhal.

Medical Imaging with Deep Learning (MIDL), 2026. [\[Paper\]](#)

Projects

Automatic Discovery and Mapping of Streetlights from Street View Videos

Project Intern | [\[Report\]](#) | [\[Video\]](#)

iHub-Data, India

August 2022 – March 2023

- Developed GeoLighting, an end-to-end, three-stage computer vision pipeline for automatic discovery and geo-localization of streetlights from GPS-tagged street-view videos, enabling scalable streetlight map generation.
- Used YOLOv8n for object detection and ORB-based feature matching to identify and track unique streetlights across video frames.
- Achieved 98.7% mAP@0.5 for detection and an average localization error of 1.3–2.7 m across diverse environments.

Incipient Technologies

Data Science Intern

India

June 2021 – August 2021

- Developed a smart surveillance system for traffic applications involving object counting, license plate recognition using OCR, and multi-object tracking in low-resolution video feeds.
- Experimented with object detection and tracking models to develop an optimized architecture, which was integrated with diverse CCTV cameras for near-real-time inference.

Age and Gender Detection for a Low-Resolution Video Surveillance Feed

Inter IIT Tech Meet 10.0, Bosch AI | [\[Slides\]](#) | [\[Video\]](#)

India

Feb 2022 – April 2022

- Worked on building a Real-Time Inference of Age and Gender Detection novel model for a Low-Resolution real-time video camera surveillance.
- Built a novel six-stage deep learning pipeline with Super Resolution and GAIT Sequencing for efficient and robust age and gender detection of the crowd.

Domain Generalization and Scene Text Detection

Project Intern | [\[Material\]](#)

Multimedia Lab, IIT (BHU)

May 2022 – July 2022

- Enhanced robustness of object detection models for both in-distribution and out-of-distribution scenarios by leveraging synthetic data generation and advanced domain adaptation/generalization training strategies.

- Successfully implemented and optimized the Progressive Contour Regression for Arbitrary-Shape Scene Text Detection (Deep Snake) approach, improving performance and training efficiency through continual learning techniques such as curriculum, progressive, and pruning-based learning.

Laboratory of Information Photonics & Optical Metrology

IIT (BHU), India

Thesis Project | Advisor: [Dr. Rakesh](#)

Jan 2022 – Dec 2022

- Investigated a non-interferometric Stokes-parameter correlation approach for quantitatively determining the topological charge and phase structure of vortex beams disrupted by random scattering media through theoretical, experimental, and MATLAB-based analysis. | [\[Thesis\]](#)
- Implemented and investigated a deep learning-based approach for real-time 3D field imaging using holography, leveraging the DeepCGH package and synthetic as well as experimental holograms. | [\[Project\]](#)

Network Analysis

IIT (BHU), India

Course Project | [\[Slides\]](#)

Aug 2023 – Nov 2023

- Analyzed and interpreted protein interaction networks using Gephi and Python to characterize network structure, topology, and scale-free properties.
- Identified communities, connected components, and other key structural characteristics through comprehensive network analysis.

Achievements

- Recipient of [DAAD-WISE Fellowship](#), a research-oriented scholarship program by the German Government, 2023
- Recipient of [Summer Research Fellowship](#), a research program by the Indian Government, 2023
- Secured an All India Rank under 8000 out of 1.2 million candidates in JEE Advanced, 2019
- Ranked within the Top 1% of students pursuing higher secondary education in the Maharashtra Secondary and Higher Secondary Board Examinations, 2019

Skills and Areas of Interest

Programming: Python, C, C++, MATLAB, SQL, LaTeX

ML/AI Libraries: PyTorch, TensorFlow, Transformers, Timm, OpenCV, NumPy, SciPy, Pandas, Matplotlib, Scikit-Learn

Miscellaneous: DeepCGH, Docker, Heroku, 3D Slicer, Gephi, Quantum Espresso

Areas of Interest: Medical AI, Multiple Instance Learning, Generative AI, OOD Generalization, Histopathology Foundation Models, Explainable AI

Positions of Responsibility

- Teaching Assistant (TA), PHY 101 - Classical, Quantum & Relativistic Mechanics; 2024
- Core Member, Business Club, IIT Varanasi; 2020 - 2021

Relevant Coursework

Computer Vision, Machine Learning and Optimization (*), Applied Deep Learning, Data Mining, Calculus, Linear Algebra, Differential Equations, Discrete Mathematics (*), Computer Programming, Graph Theory and its Applications, Statistical Physics, Computational Physics, Quantum Physics, Analogue Circuits and Systems, Digital Circuits and Systems, Quantum Electronics, Solid State Electronic Devices, Fibre Optics, Instrument Measurement and Analysis

* - MOOCs